SFT initialization: start with plain, long CoT, and then additional
⇒ claimed benefit: interpretability ⇒ generate well-formatted verification CoT.
⇒ Note: origins on the data not quite clear...

SFT for Reasoning / math:
⇒ for reasoning/math ability
A small number of samples for bootstrapping Reasoning from LLMs.

RL steps: RL part is same for dpsk-R1-zero and dpsk-R1
for longer CoT.
Minor difference: additional language consistency Rewards
~~Interesting~~ Cons: RL ~~naturally~~ Sometimes leads to mixed languages

SFT / RLHF: the usual post-train process happens after RLVR / Reasoning RL
for alignment

~~Step~~ SFT step, 2 epoches:
dpsk: post-train
① Reasoning data → non-verifiable tasks, use Vs as a judge
② non-Reasoning data - V3 SFT dataset (200K)

RLHF step: alignment
① Re-use R1-zero style reasoning RLHF
② non-verifiable tasks - V3 RLHF pipeline, still use GRPO for RLHF

How well does R1 work? : pretty good

Distillation: can we get non-reasoning models to reason
Pipeline: ① Have R1 generate ⇒ CoT traces (800K) data
② SFT teach Qwen 2.5 via distillation ↘ SFT

insights for supervised distillation:

①: small language model can obtain significant reasoning ability

②: Supervised distillation is better than RL for post-training small language model

③: open-source: dpsk-R1 → 800k CoT trace data → SFT Qwen2.5

{ coding, math, scientific discovery } ← Reasoning-Qwen2.5 ←

Kimi-k1.5: why do we study this?

①: Released at the same time as R1

②: also beats of using RL.

③: scaling Reinforcement learning with LLMs

④: Technical Report for KIMI-1.5

⇒

pros: Long COT Reasoning strategy: key steps. filtering

①: Dataset Conseruation (difficulty ~~conseruation~~)

②: SFT: long COT

③: RL: with their own policy gradient loss

Data curation / SFT: data, math-style setting, balancing topics

{ exclude multiple choice / True False
select only examples that models fail on

DPO-type derivation for RL

{ Baselined policy gradient w/ regularization = like GRPO clipping
Reference_based Reward model

best-of-8